

Accelerating Multi-Target Regressor Chains via Joint L_1 -Norm Weight Magnitude Dependency Ordering

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Abstract—Multi-Target Regressor Chains (MRC) exploit inter-target dependencies by modeling target sequences where preceding predictions act as features for subsequent models. However, selecting the optimal target sequence traditionally relies on an iterative greedy search based on Mean Squared Error (MSE), exhibiting a computational complexity of $\mathcal{O}(K^2)$ with respect to target count K . In this paper, we propose a single-pass $\mathcal{O}(1)$ heuristic target selection method leveraging L_1 -norm weight magnitude summation ($\sum |w_{ij}|$) over a robustly scaled joint linear space solved via SVD pseudo-inverse. Evaluated across synthetic and real-world datasets, the proposed formulation achieves up to a $725.11\times$ speedup over the iterative greedy baseline while maintaining exact predictive MSE parity (Zero Accuracy Loss).

Index Terms—Multi-Target Regression, Regressor Chains, Algorithm Optimization, Linear Algebra, Pseudo-Inverse, L_1 -Norm Summation.

I. INTRODUCTION

Multi-target regression involves predicting multiple continuous outputs given a set of input features. Multi-Target Regressor Chains (MRC) significantly improve prediction performance over independent regressors by chaining targets sequentially. Despite its effectiveness, target sequence ordering remains a critical computational bottleneck. Finding the optimal sequence using exhaustive or greedy search strategies requires fitting K^2 candidate models, rendering the approach computationally expensive for large-scale applications.

We address this limitation by introducing a joint feature-target dependency mapping heuristic. Instead of training multiple iterative models to measure residual error, we quantify feature-target connectivity strength via an L_1 -norm weight magnitude summation ($\|W_{:,j}\|_1$) on a single-pass linear fit.

II. MATHEMATICAL FORMULATION

The optimization pipeline follows a mathematically rigorous eight-stage framework:

A. Robust Feature and Target Scaling

To eliminate scale disparity across target variables while maintaining robustness against outliers, feature matrix $X \in \mathbb{R}^{N \times M}$ and target matrix $Y \in \mathbb{R}^{N \times K}$ are transformed using the Interquartile Range (IQR):

$$IQR_j = Q_3(Y_{:,j}) - Q_1(Y_{:,j}) \quad (1)$$

$$X_{\text{scaled}} = \frac{X - \text{median}(X)}{IQR_X}, \quad Y_{\text{scaled}} = \frac{Y - \text{median}(Y)}{IQR_Y} \quad (2)$$

B. Joint Linear System Decomposition

The linear mapping between normalized inputs and targets is represented as:

$$X_{\text{scaled}} W \approx Y_{\text{scaled}} \quad (3)$$

Using Singular Value Decomposition (SVD) where $X = U\Sigma V^T$, the Moore-Penrose Pseudo-Inverse X_{pinv} is computed:

$$X_{\text{pinv}} = (X^T X)^{-1} X^T = V \Sigma^+ U^T \quad (4)$$

The joint optimal weight matrix $W \in \mathbb{R}^{M \times K}$ is obtained in a single operation:

$$W = X_{\text{pinv}} Y_{\text{scaled}} \quad (5)$$

C. Target Dependency Score (L_1 -Norm Summation)

The core heuristic quantifies overall input dependency strength per target column j by summing absolute coefficient magnitudes:

$$S_j = \|W_{:,j}\|_1 = \sum_{i=1}^M |w_{ij}| \quad (6)$$

D. Sequence Permutation and Model Fitting

Target indices are sorted in descending order of their L_1 -norm scores to yield the optimal chain order π :

$$\pi = \text{argsort}(\mathbf{S})_{\text{descending}} = [j_{(1)}, j_{(2)}, \dots, j_{(K)}] \quad (7)$$

The Regressor Chain is then trained sequentially where target $j_{(k)}$ receives input features combined with predictions from previous targets $[X, Y_{:,j_{(1)}}, \dots, Y_{:,j_{(k-1)}}]$.

III. EMPIRICAL EVALUATION

The proposed methodology was benchmarked against the standard Iterative Greedy MSE Search on both synthetic datasets and the real-world California Housing dataset.

As detailed in Table I, the selection runtime of the proposed approach on 80,000 samples with 20 targets drops from 48.21 seconds to just 66.49 milliseconds while maintaining exact prediction accuracy parity.

TABLE I
PERFORMANCE COMPARISON: GREEDY BASELINE VS. PROPOSED
METHOD

Dataset	Rows	Greedy (ms)	O(1) (ms)	Speedup	MSE
Small Synth	800	90.08	33.58	$2.68\times$	0.000464
Med Synth	8,000	1,433.43	7.30	$196.33\times$	0.000464
Large Synth	80,000	48,212.92	66.49	$725.11\times$	0.000023
Cal. Housing	16,512	82.47	11.77	$7.01\times$	0.462408

IV. CONCLUSION

This study proves that an L_1 -norm weight magnitude summation over a single-pass normalized pseudoinverse matrix serves as an effective, highly scalable proxy for Multi-Target Regressor Chain sequence ordering. The approach eliminates the quadratic $O(K^2)$ model fitting loop, making high-dimensional regressor chains computationally practical for real-time systems.

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